

Total No. of Questions : 8]

SEAT No. :

P-3153

[Total No. of Pages : 2

[6003]-354

T.E. (Computer Engineering)
Data Science and Big Data Analytics
(2019 Pattern) (Semester - II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

Q1) a) What is Model Building elaborate this phase of data analytics with the help of suitable example. **[9]**

b) Explain any three sources of Big Data. Differentiate BI versus Data science. **[8]**

OR

Q2) a) What are the three characteristic of Big Data and what are the main consideration in processing Big Data. **[8]**

b) Explain Descriptive, Diagnostic, Predictive analytics. **[9]**

Q3) a) Explain why decision tree are used. Draw a sample decision tree and explain its parts. **[9]**

b) How Apriori Algorithm works, explain with suitable example? **[9]**

OR

Q4) a) What is data preprocessing? Explain in details about handling missing data and transformation of data. **[9]**

b) Explain Naïve Bayes' classifier and it applications. **[9]**

P.T.O.

- Q5) a)** What is text processing? Explain TF-IDF with example. [8]
b) With suitable example, explain the steps involved in k-means algorithm. [9]

OR

- Q6) a)** Define following terms with respect to confusion matrix : [8]
i) Accuracy
ii) Precision
iii) Recall
iv) AUC-ROC
b) Explain k-fold Cross Validation & Random Subsampling. [9]

- Q7) a)** With a suitable example, draw a Histogram, boxplot and explain its usages. [9]
b) Describe the data visualization tool Tableau. List of data visualization tools. [9]

OR

- Q8) a)** What is Data Visualization? Describe the challenges of data visualization. [9]
b) Explain architecture of Apache-Pig. [9]

